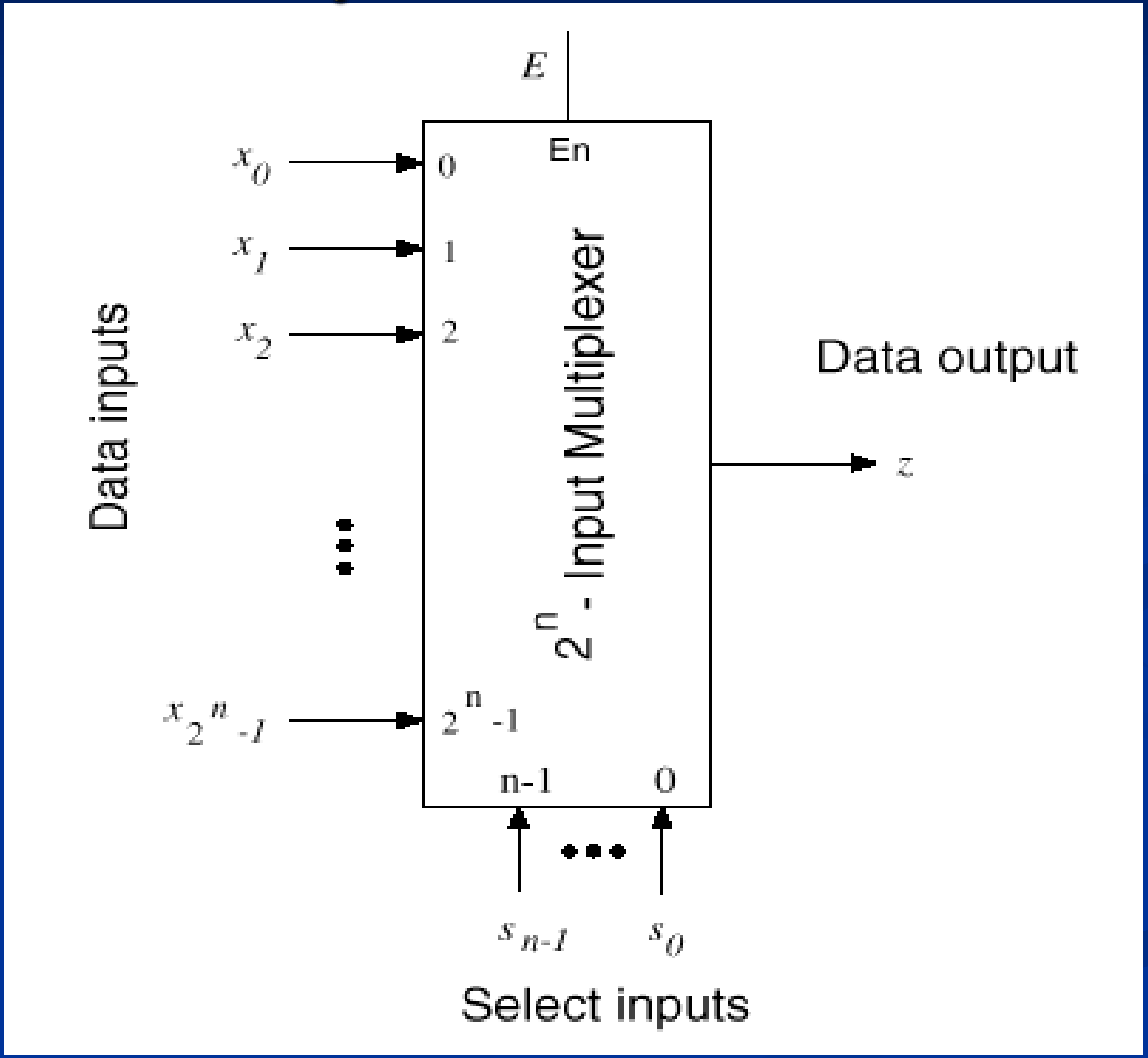
Overview of Encoder and Decoder

- MUX Gate
- Rudimentary functions
- Binary Decoders
  - Expansion
  - Circuit implementation
- Binary Encoders
- Priority Encoders

Multiplexer

- "Selects" binary information from one of many input lines and directs it to a single output line.
- Also know as the "selector" circuit,
- Selection is controlled by a particular set of inputs lines whose # depends on the # of the data input lines.
- For a 2^n-to-1 multiplexer, there are 2^n data input lines and *n* selection lines whose bit combination determines which input is selected.

Multiplexer (cont.)

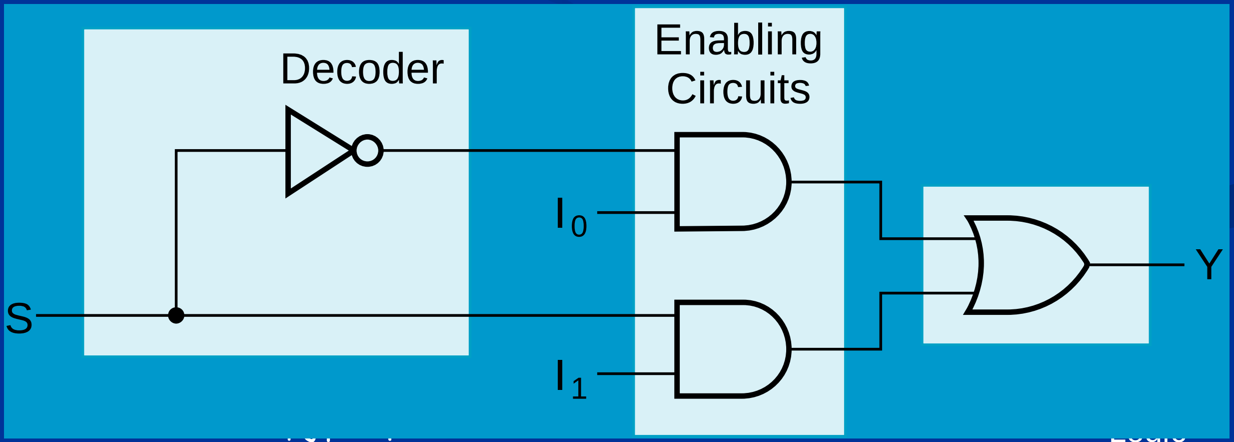


2-to-1-Line Multiplexer

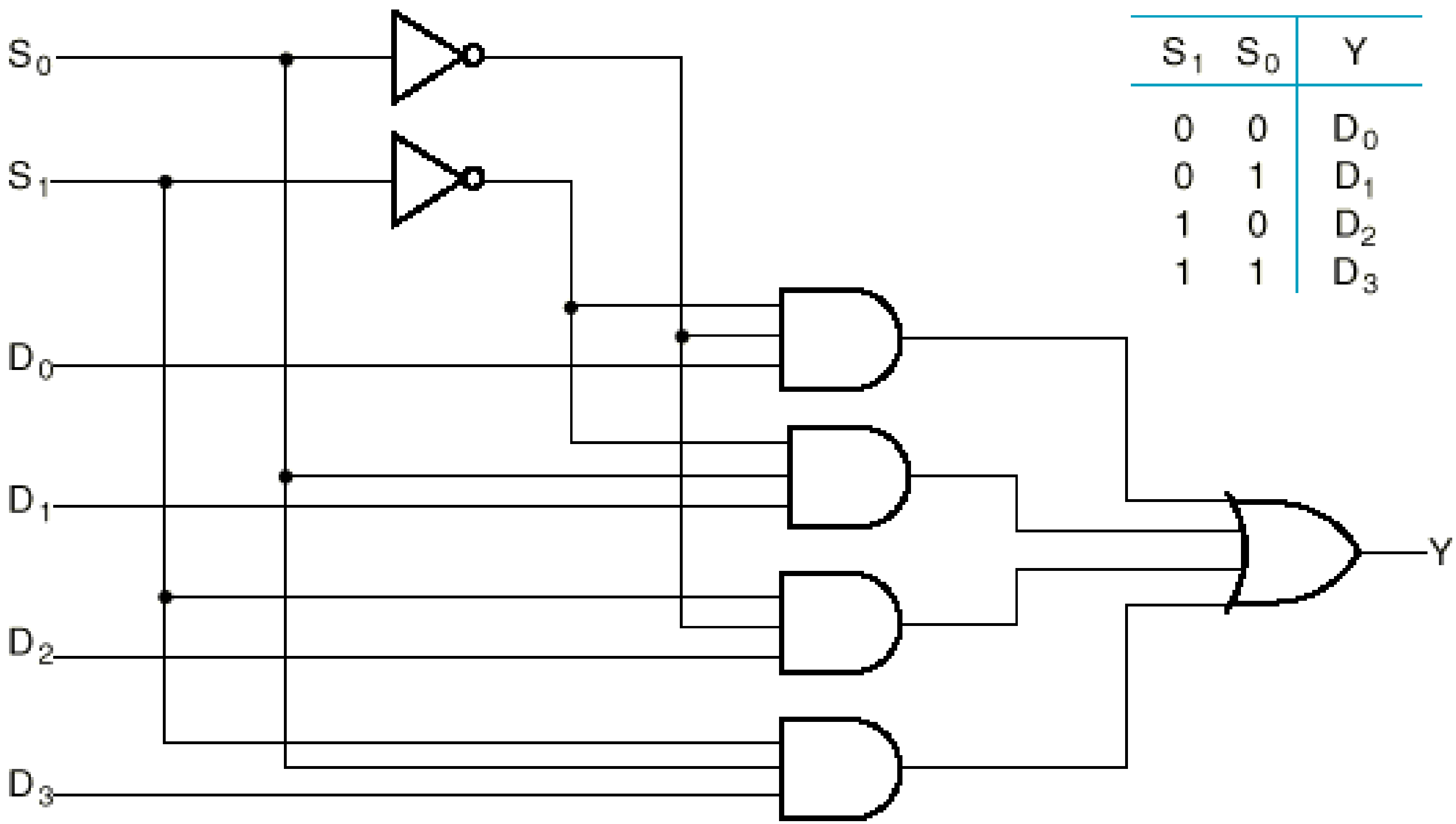
- Since $2 = 2^1$, $n = 1$
- The single selection variable S has two values:
  - S = 0 selects input I_0
  - S = 1 selects input I_1
- The equation:

$$Y = S' I_0 + S I_1$$

- The circuit:



Example: 4-to-1 MUX using Cell Library Based Design

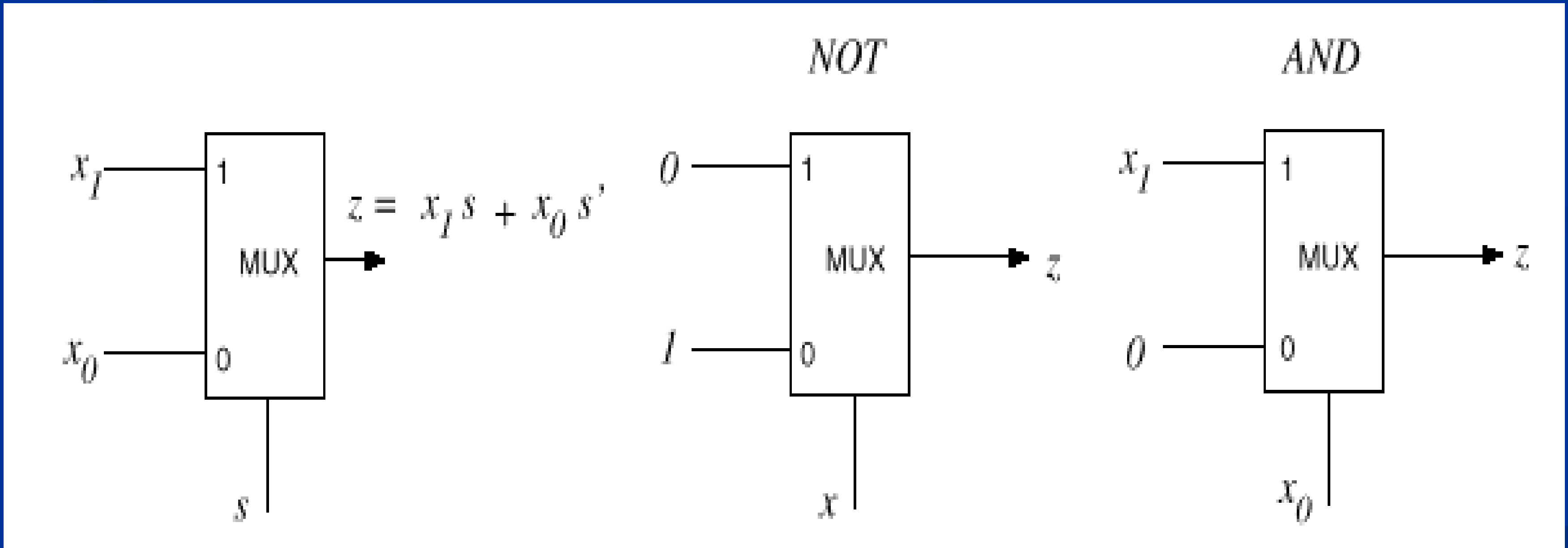


Function table

S_1	S_0	Y
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

MUX as a Universal Gate

- We can construct AND and NOT gates using 2-to-1 MUXs. Thus, 2-to-1 MUX is a universal gate.



$z = 0x + 1x' = x'$

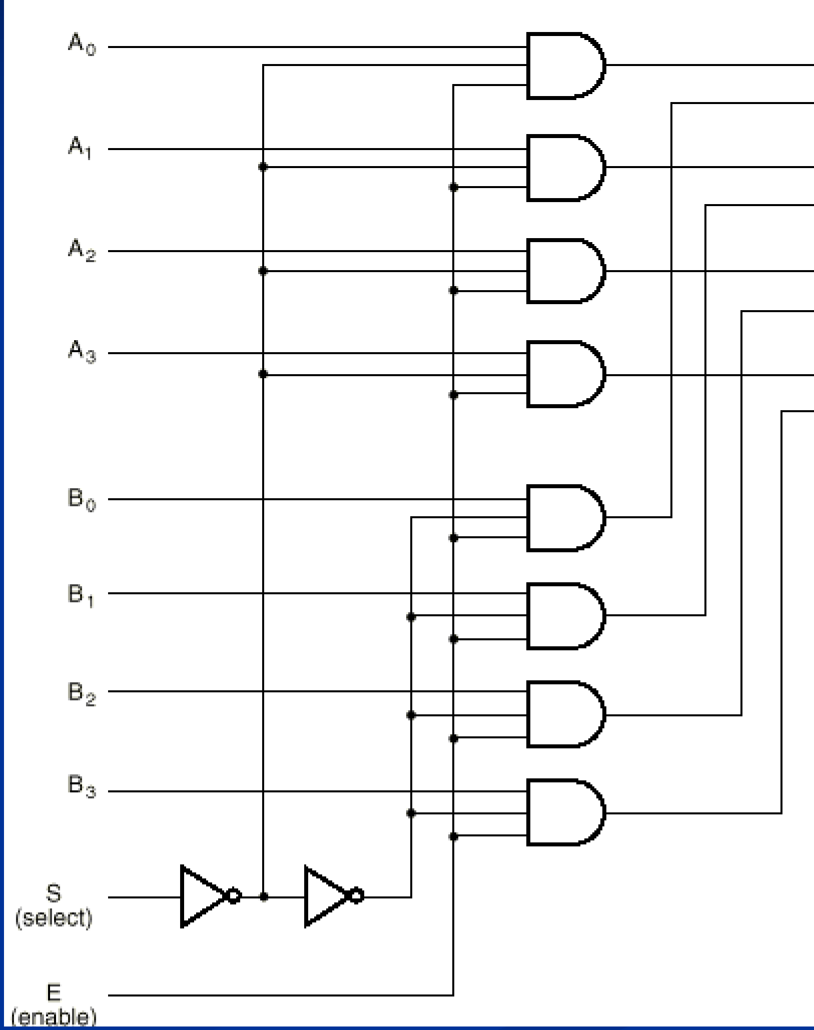
$z = x_1x_0 + 0x_0' = x_1x_0$

Multiple Bit Selection

- Until now, we have examined single-bit data selected by a MUX. What if we want to select m-bit data/words?
→ Combine MUX blocks in parallel with common select and enable signals
- Example: Construct a logic circuit that selects between 2 sets of 4-bit inputs (see next slide for solution).

Example: Quad 2-to-1 MUX

- Uses four 4-to-1 MUXs with common select (S) and enable (E).
- Select line chooses between A_i's and B_i's. The selected four-wire digital signal is sent to the Y_i's
- Enable line turns MUX on and off (E=1 is on).



Function table

E	S	Output Y
0	X	All 0's
1	0	Select A
1	1	Select B

Implementing Boolean functions with Multiplexers

- Any Boolean function of *n* variables can be implemented using a 2^{n-1}-to-1 multiplexer. A MUX is basically a decoder with outputs ORed together, hence this isn't surprising.
- The SELECT signals generate the minterms of the function.
- The data inputs identify which minterms are to be combined with an OR.

Efficient Method for implementing Boolean functions

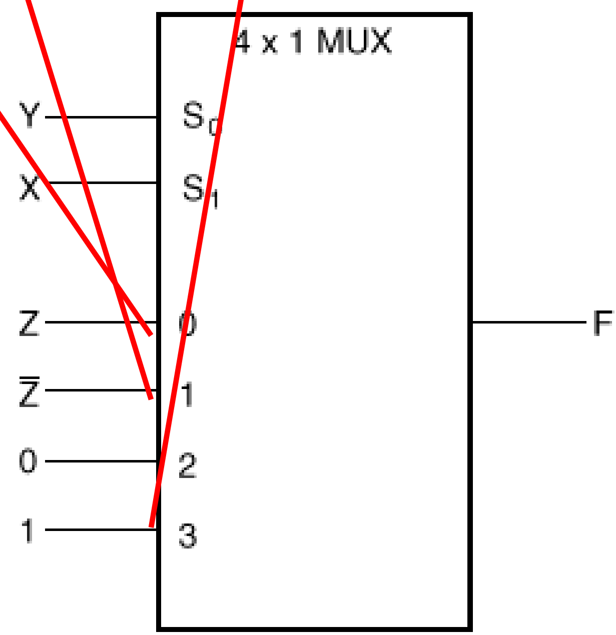
- For an *n*-variable function (*e.g.*, f(A,B,C,D)):
  - Need a 2^{n-1} line MUX with *n*-1 select lines.
  - Enumerate function as a truth table with consistent ordering of variables (*e.g.*, A,B,C,D)
  - Attach the most significant *n*-1 variables to the *n*-1 select lines (*e.g.*, A,B,C)
  - Examine pairs of adjacent rows (only the least significant variable differs, *e.g.*, D=0 and D=1).
  - Determine whether the function output for the (A,B,C,0) and (A,B,C,1) combination is (0,0), (0,1), (1,0), or (1,1).
  - Attach 0, D, D', or 1 to the data input corresponding to (A,B,C) respectively.

Example

- $F(X,Y,Z) = X'Y'Z + X'YZ' + XYZ' + XYZ = \Sigma m(1,2,6,7)$
- There are n=3 inputs, thus we need a 2^2-to-1 MUX
- The first n-1 (=2) inputs serve as the selection lines

X	Y	Z	F	
0	0	0	0	F = Z
0	0	1	1	
0	1	0	1	F = $\overline{Z}$
0	1	1	0	
1	0	0	0	F = 0
1	0	1	0	
1	1	0	1	F = 1
1	1	1	1	

(a) Truth table



(b) Multiplexer implementation

The Other Example

- Consider $F(A,B,C) = \sum m(1,3,5,6)$. We can implement this function using a 4-to-1 MUX as follows.
- The index is ABC. Apply A and B to the S_1 and S_0 selection inputs of the MUX (A is most sig, S_1 is most sig.)
- Enumerate function in a truth table.

MUX Example (cont.)

When A=B=0, F=C

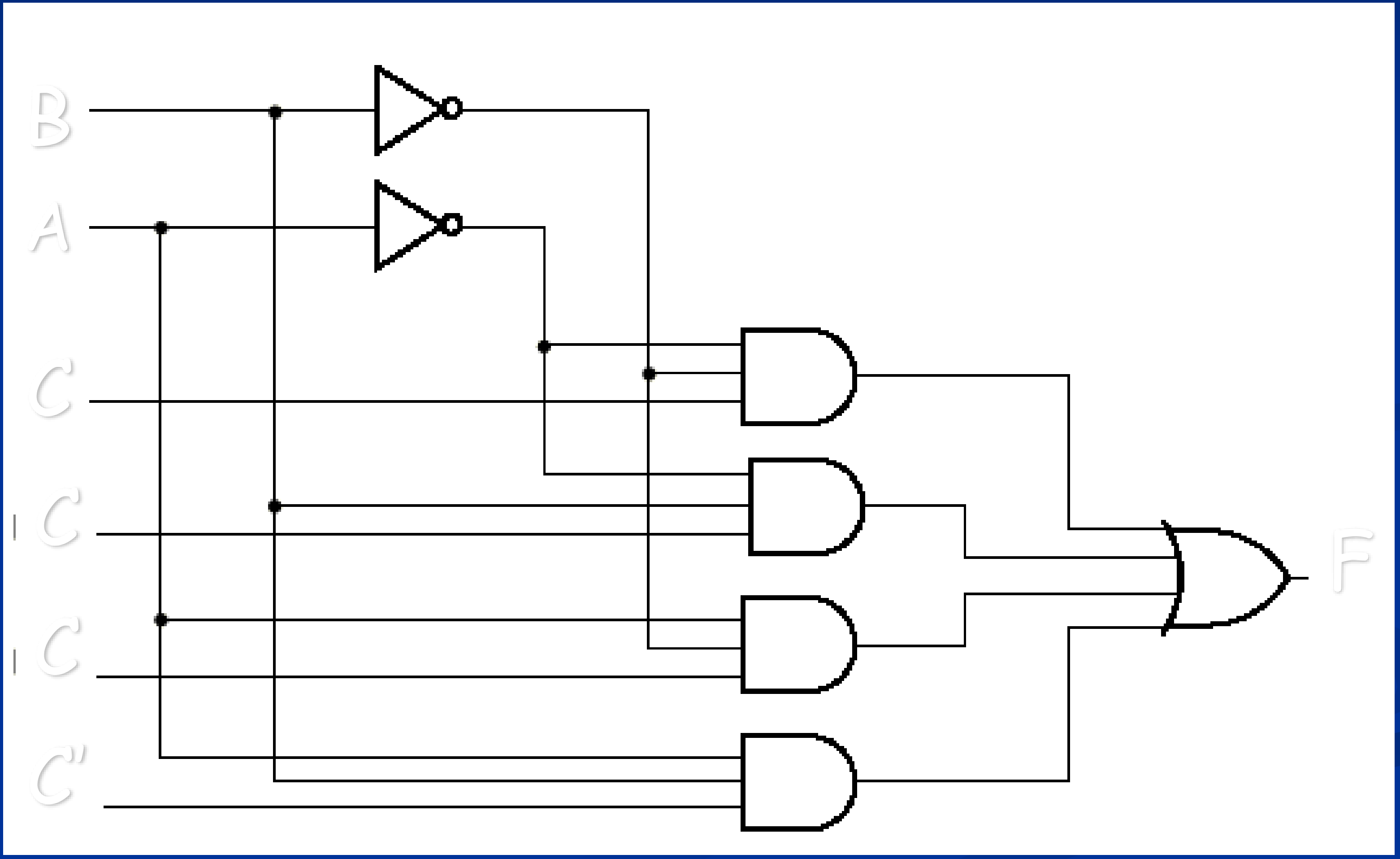
When A=0, B=1, F=C

When A=1, B=0, F=C

When A=B=1, F=C'

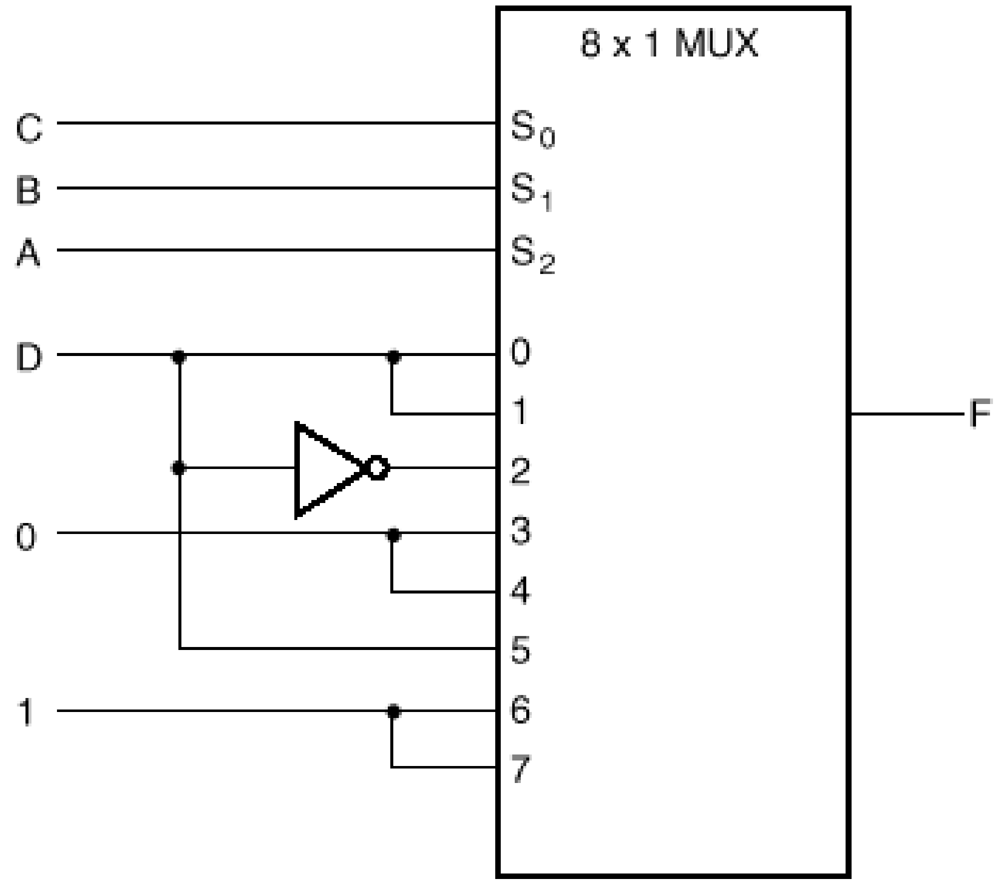
A	B	C		F
0	0	0		0
0	0	1		1
0	1	0		0
0	1	1		1
1	0	0		0
1	0	1		1
1	1	0		1
1	1	1		0

MUX implementation of F(A,B,C) = $\sum m(1,3,5,6)$



A larger Example

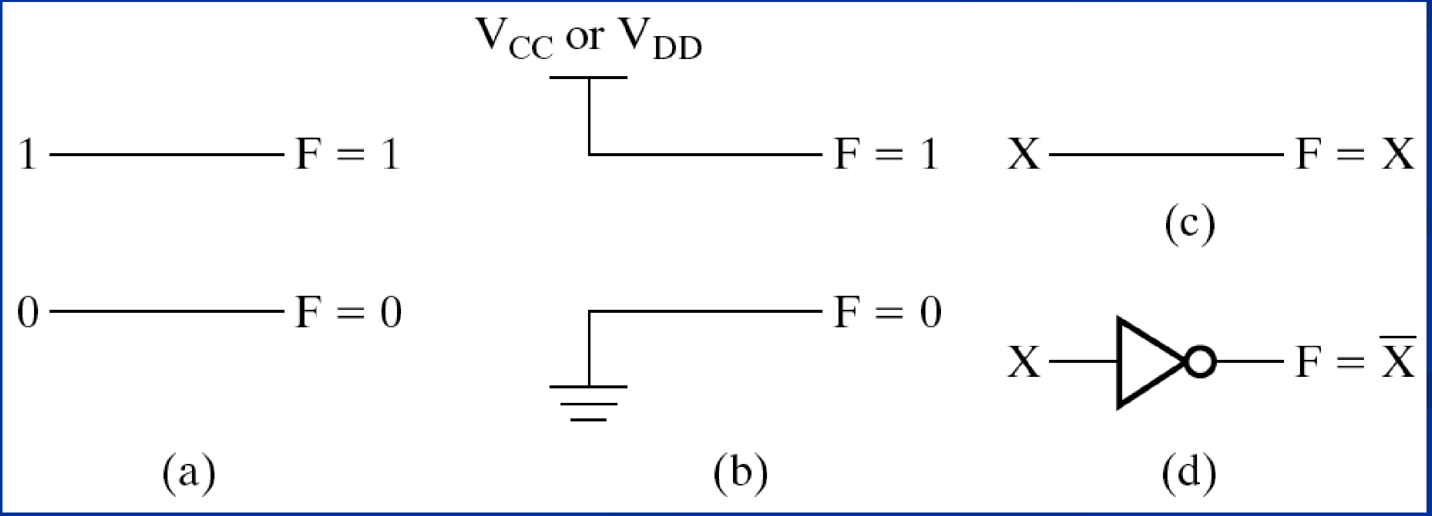
A	B	C	D	F	
0	0	0	0	0	F = D
0	0	0	1	1	
0	0	1	0	0	F = D
0	0	1	1	1	
0	1	0	0	1	F = $\overline{D}$
0	1	0	1	0	
0	1	1	0	0	F = 0
0	1	1	1	0	
1	0	0	0	0	F = 0
1	0	0	1	0	
1	0	1	0	0	F = D
1	0	1	1	1	
1	1	0	0	1	F = 1
1	1	0	1	1	
1	1	1	0	1	F = 1
1	1	1	1	1	



Rudimentary Functions

TABLE 4-1
Functions of One Variable

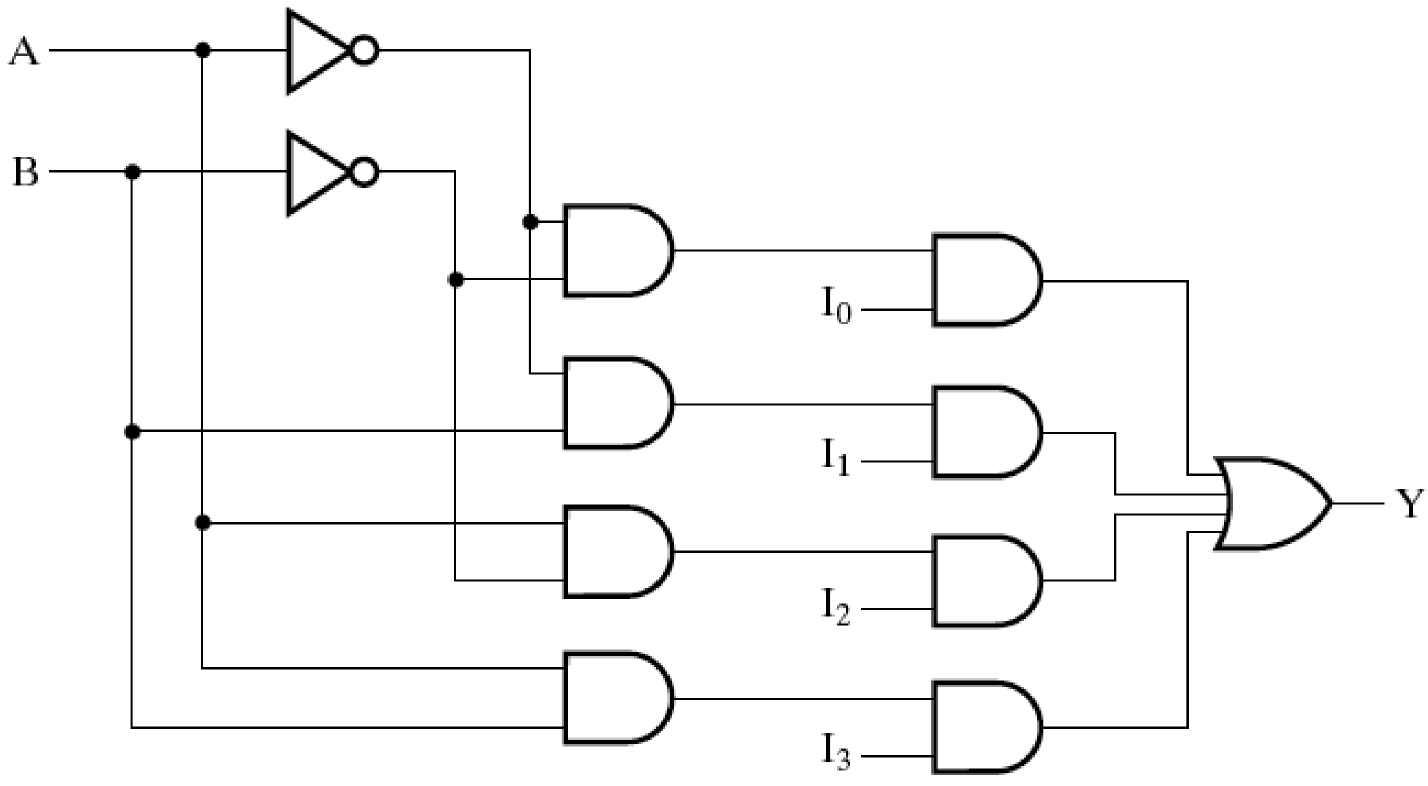
X	F = 0	F = X	$F = \bar{X}$	F = 1
0	0	0	1	1
1	0	1	0	1



Selection

A	B	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

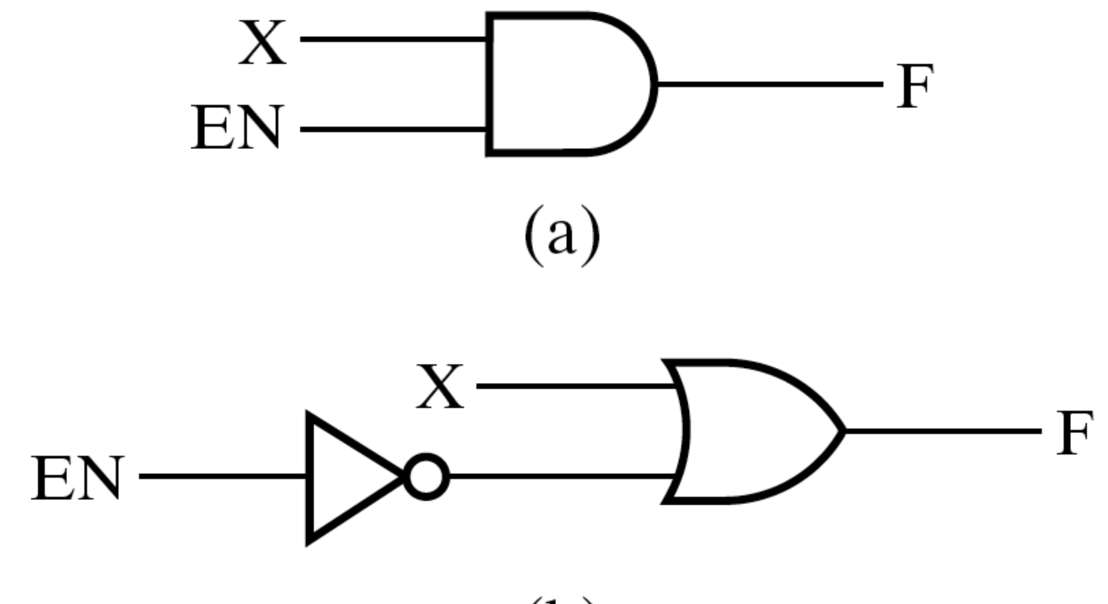
(a)



(b)

Enabling

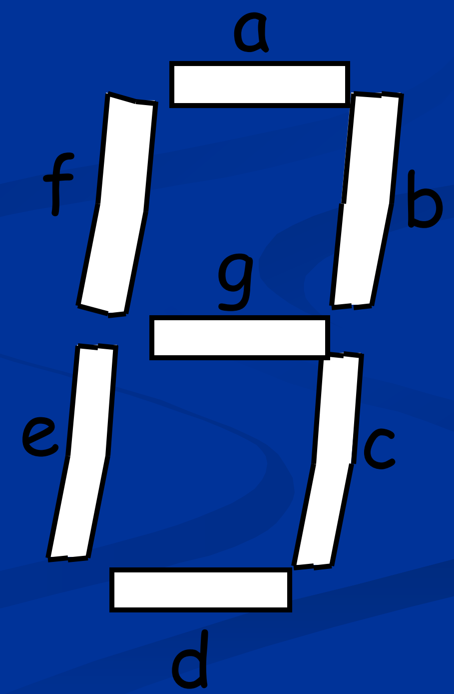
- "gating" ?



(a)
(b)

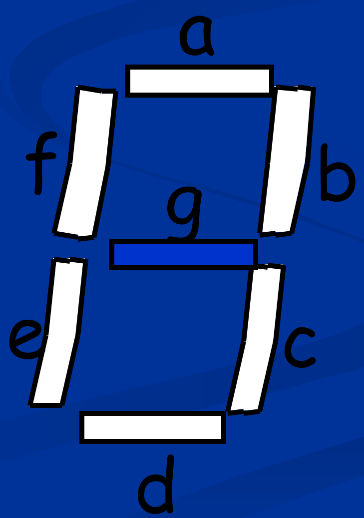
The Other Code Converter
BCD-to-Seven-Segment Converter

- Seven-segment display:
  - 7 LEDs (light emitting diodes), each one controlled by an input
  - 1 means "on", 0 means "off"
  - Display digit "3"?
    - Set a, b, c, d, g to 1
    - Set e, f to 0



BCD-to-Seven-Segment Converter

- Input is a 4-bit BCD code → 4 inputs (w, x, y, z).
- Output is a 7-bit code (a,b,c,d,e,f,g) that allows for the decimal equivalent to be displayed.
- Example:
  - Input: 0000_{BCD}
  - Output: 1111110 (a=b=c=d=e=f=1, g=0)



BCD-to-Seven-Segment (cont.) Truth Table

Digit	wxyz	abcdefg
0	0000	1111110
1	0001	0110000
2	0010	1101101
3	0011	1111001
4	0100	0110011
5	0101	1011011
6	0110	X011111
7	0111	11100X0

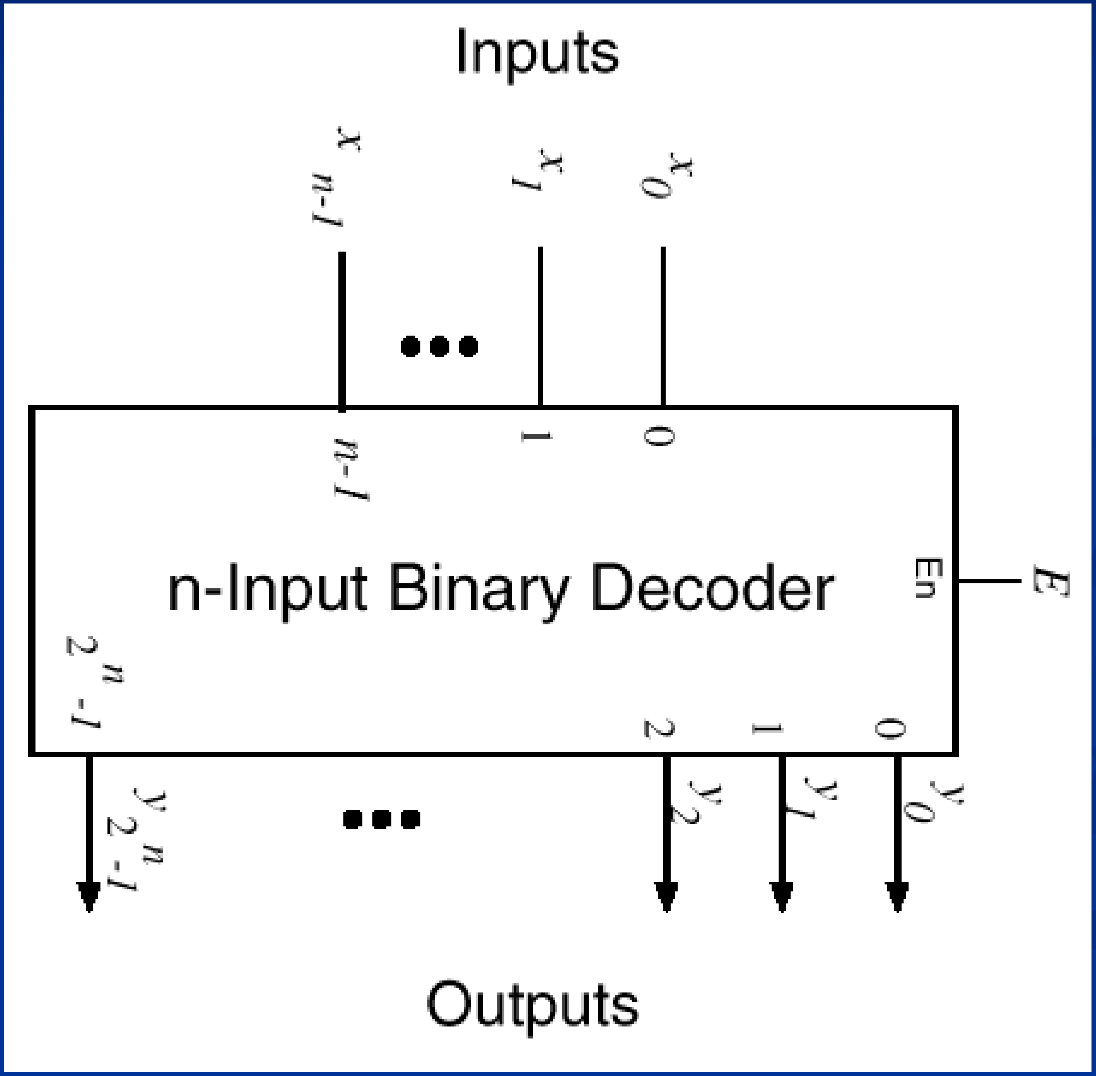
Digit	wxyz	abcdefg
8	1000	1111111
9	1001	111X011
	1010	XXXXXXX
	1011	XXXXXXX
	1100	XXXXXXX
	1101	XXXXXXX
	1110	XXXXXXX
	1111	XXXXXXX

??

Decoders

- A combinational circuit that converts binary information from *n* coded inputs to a maximum 2^n coded outputs
  → *n-to-* 2^n decoder
- *n-to-m* decoder, $m \le 2^n$
- Examples: BCD-to-7-segment decoder, where $n=4$ and $m=10$

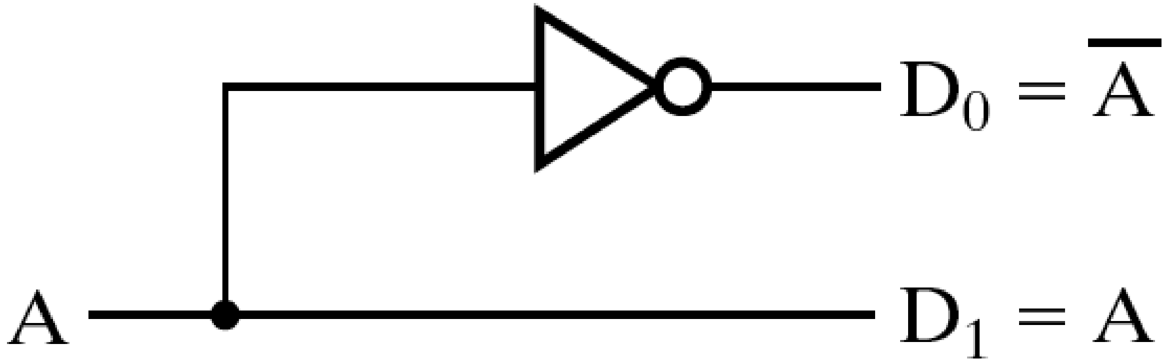
Decoders (cont.)



1-2 Decoder

A	D_0	D_1
0	1	0
1	0	1

(a)

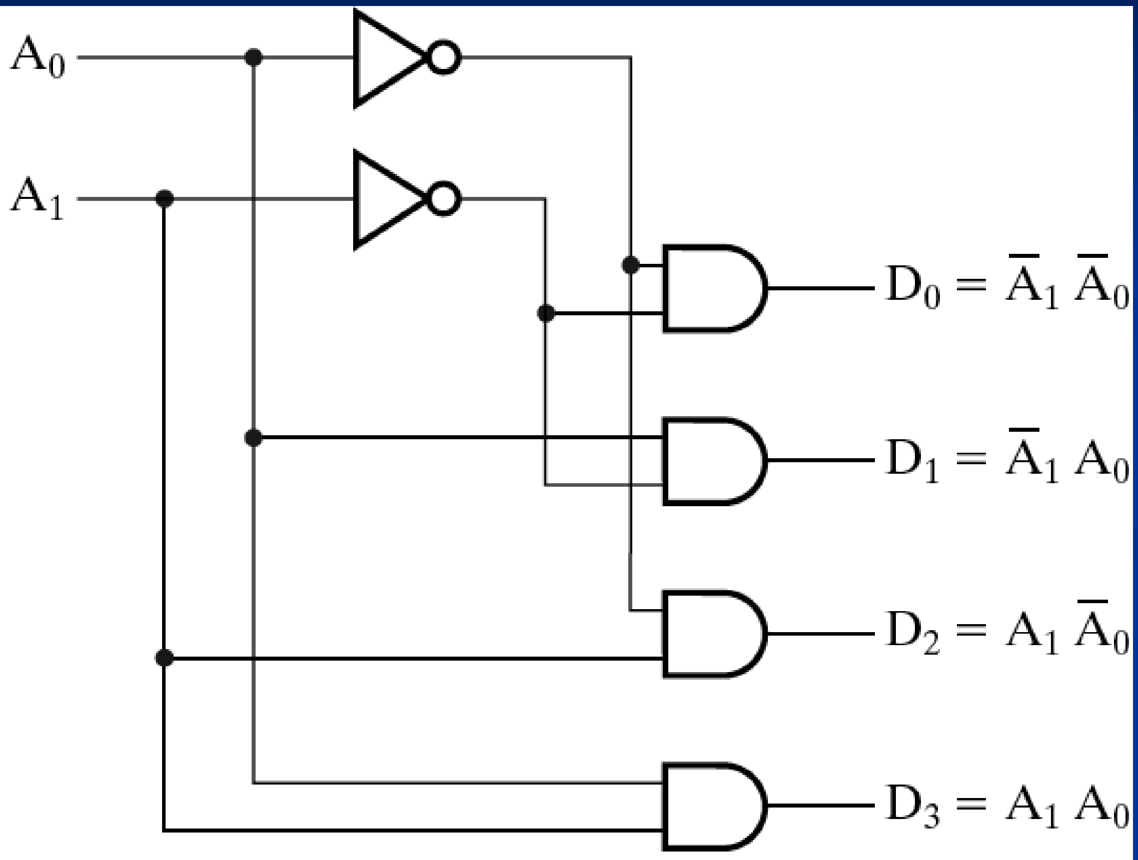


(b)

2-to-4 Decoder

A_1	A_0	D_0	D_1	D_2	D_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

(a)

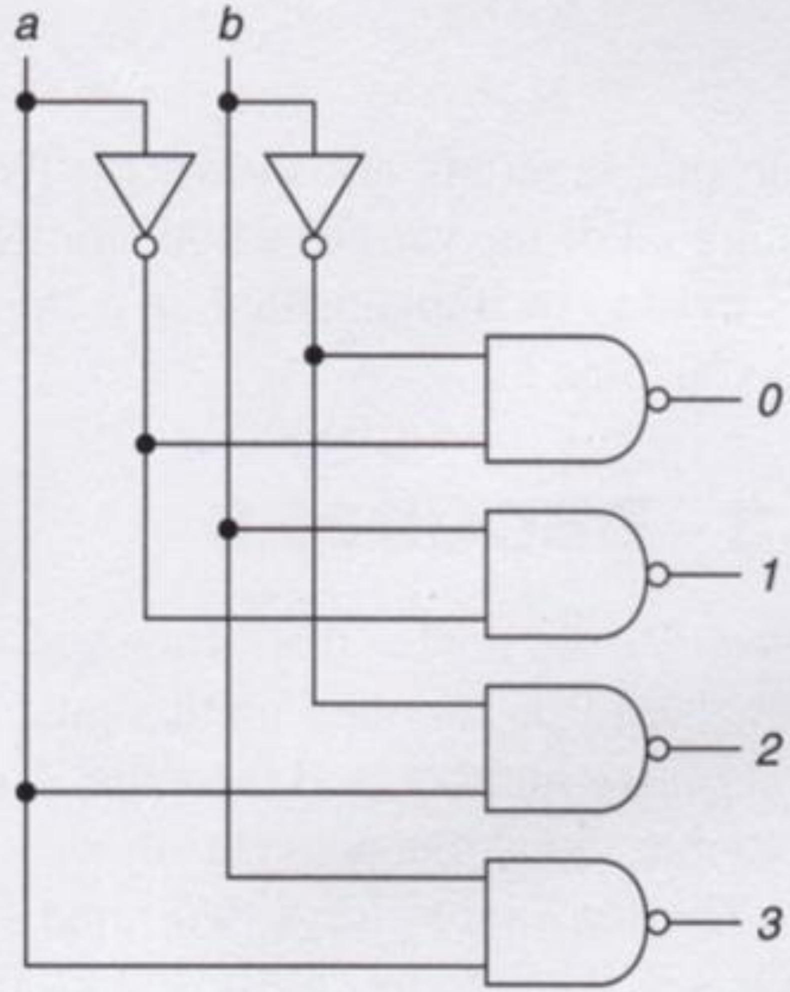


(b)

2-to-4 Active Low Decoder

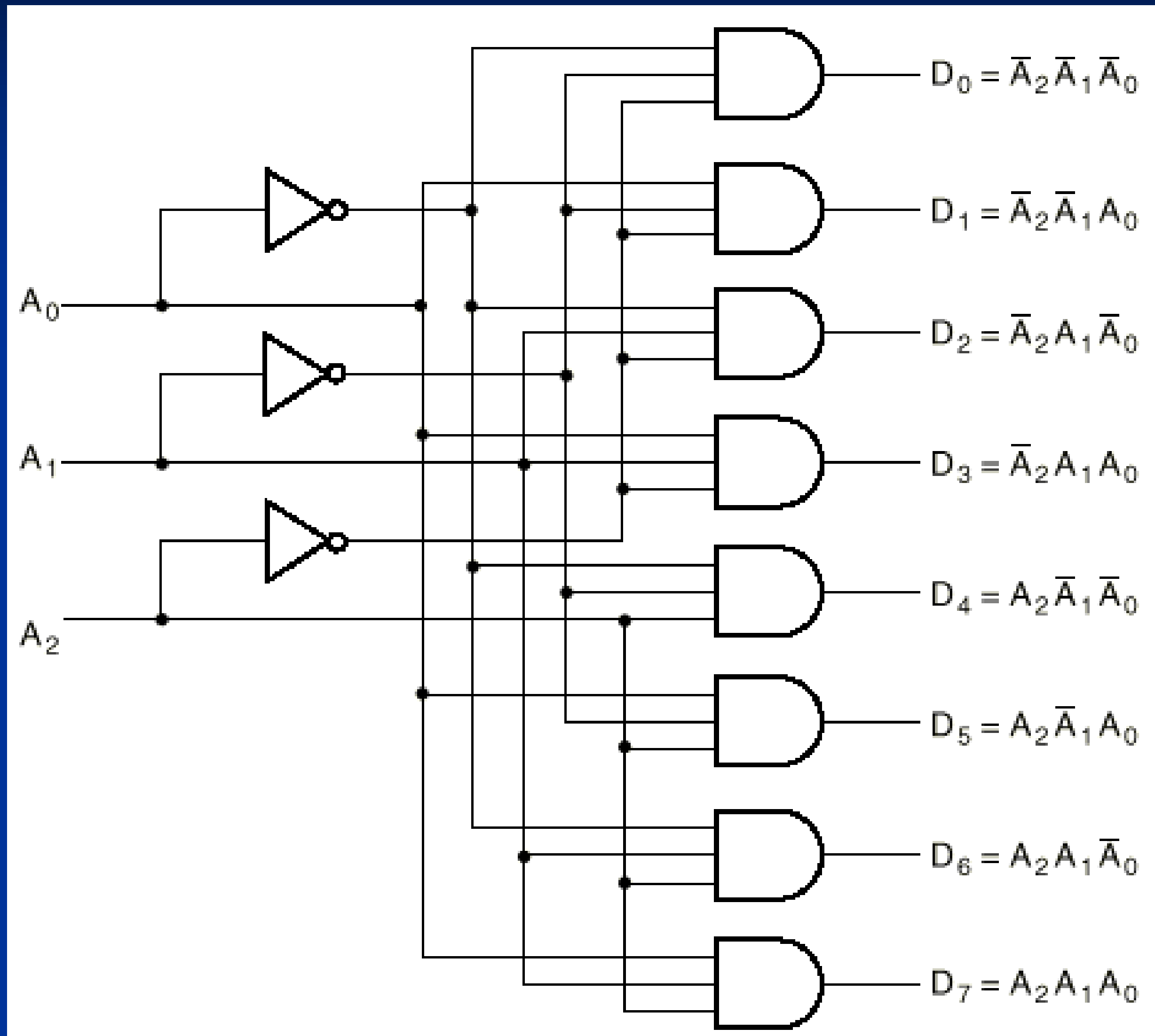
Figure 4.5 An active low decoder.

a	*b*	*0*	*1*	*2*	*3*
0	0	0	1	1	1
0	1	1	0	1	1
1	0	1	1	0	1
1	1	1	1	1	0



3-to-8 Decoder

address

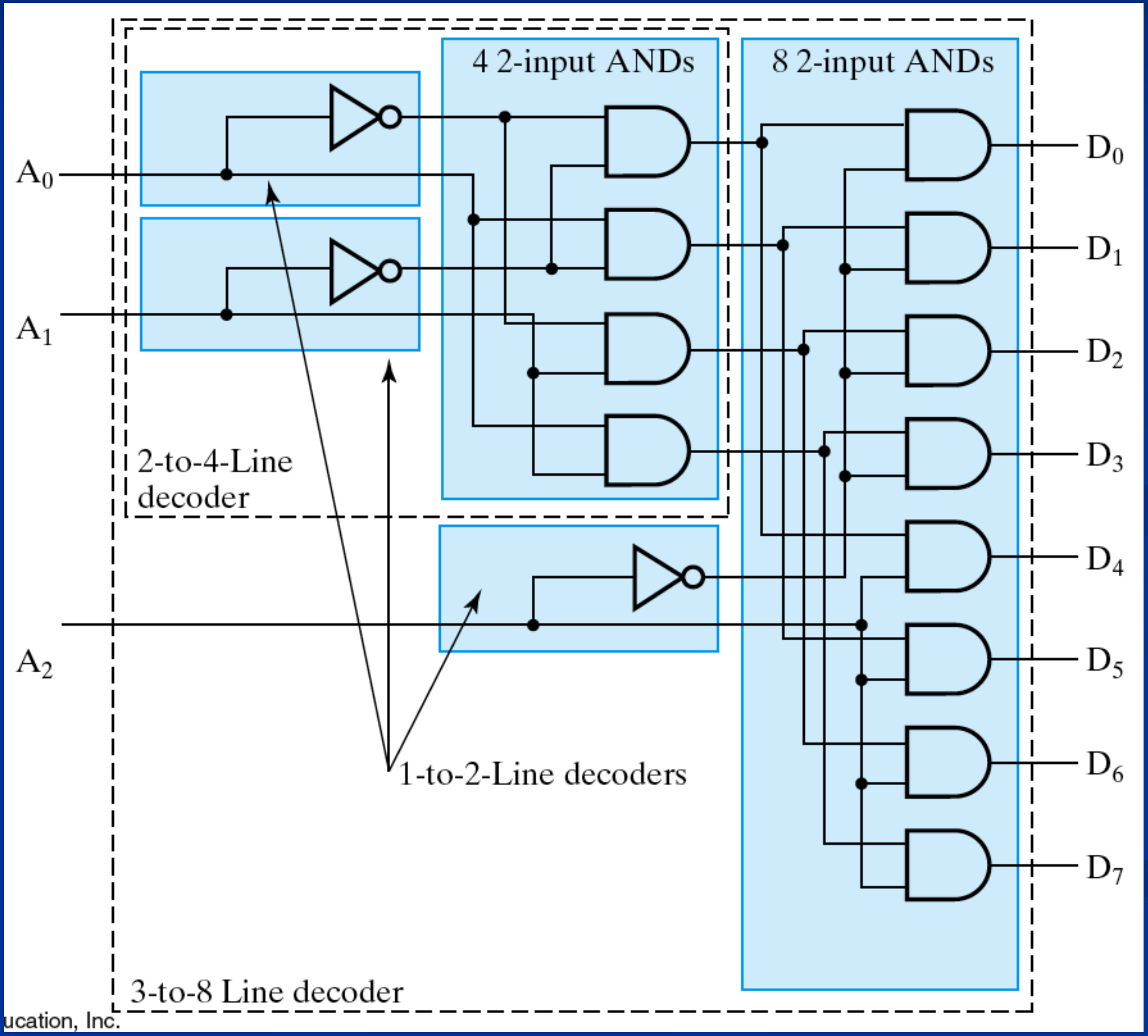


data

3-to-8 Decoder (cont.)

- Three inputs, A_0, A_1, A_2, are decoded into eight outputs, D_0 through D_7
- Each output D_i represents one of the minterms of the 3 input variables.
- $D_i = 1$ when the binary number $A_2A_1A_0 = i$
- Shorthand: $D_i = m_i$
- The output variables are *mutually exclusive*; exactly one output has the value 1 at any time, and the other seven are 0.

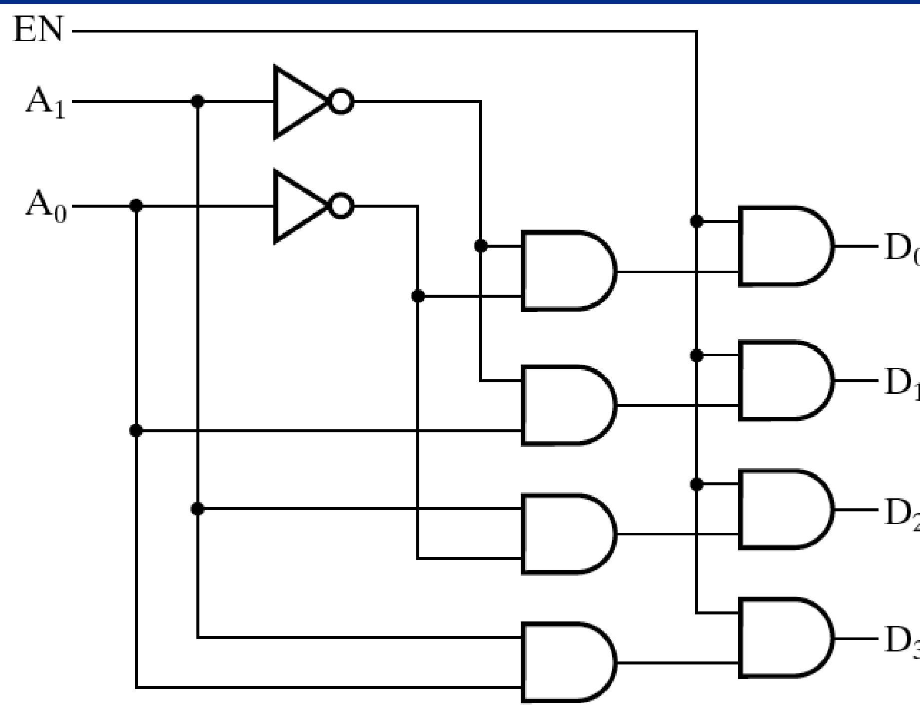
Decoder Expansion



Decoder with enable

EN	A_1	A_0	D_0	D_1	D_2	D_3
0	X	X	0	0	0	0
1	0	0	1	0	0	0
1	0	1	0	1	0	0
1	1	0	0	0	1	0
1	1	1	0	0	0	1

(a)



(b)

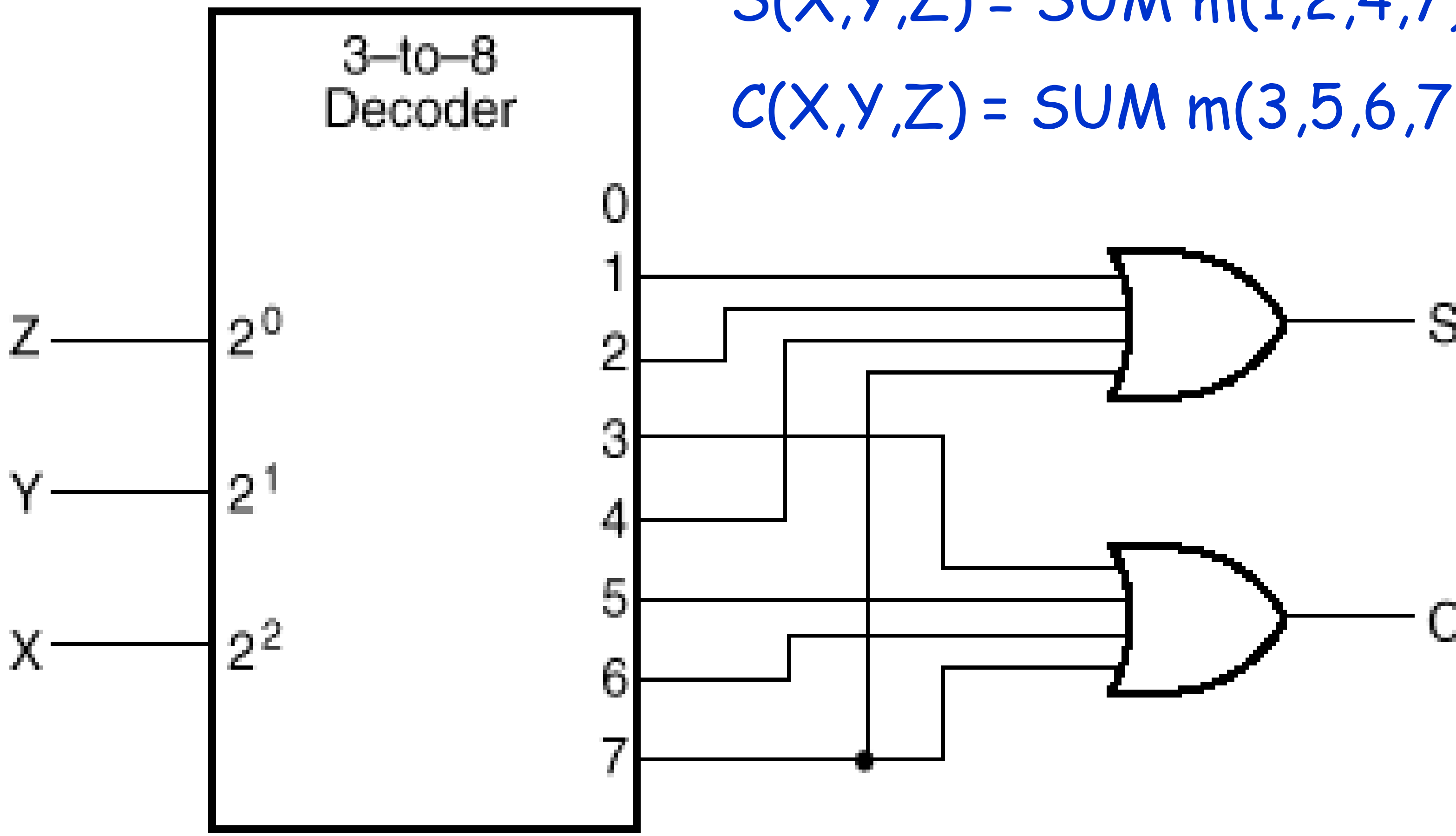
Implementing Boolean functions using decoders

- Any combinational circuit can be constructed using decoders and OR gates! Why?
- Here is an example:
  Implement a full adder circuit with a decoder and two OR gates.
- Recall full adder equations, and let X, Y, and Z be the inputs:
  - $S(X,Y,Z) = X+Y+Z = \Sigma m(1,2,4,7)$
  - $C(X,Y,Z) = \Sigma m(3, 5, 6, 7)$.
- Since there are 3 inputs and a total of 8 minterms, we need a 3-to-8 decoder.

Implementing a Binary Adder Using a Decoder

$S(X,Y,Z) = \text{SUM } m(1,2,4,7)$

$C(X,Y,Z) = \text{SUM } m(3,5,6,7)$



Encoders

- An encoder is a digital circuit that performs the inverse operation of a decoder. An encoder has 2^n input lines and n output lines.
- The output lines generate the binary equivalent to the input line whose value is 1.

Encoders (cont.)

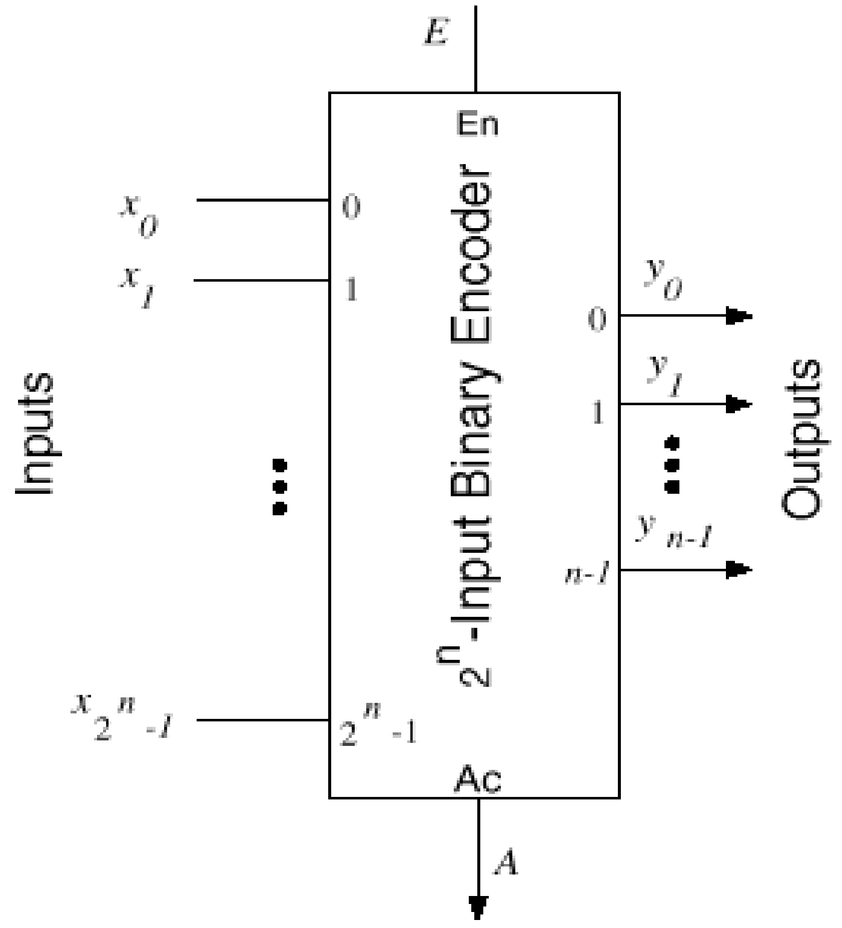


Figure 9.12: 2--input binary encoder.

Encoder Example

- Example: 8-to-3 binary encoder (octal-to-binary)

Inputs								Outputs		
D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0	A_2	A_1	A_0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

$$A_0 = D_1 + D_3 + D_5 + D_7$$
$$A_1 = D_2 + D_3 + D_6 + D_7$$
$$A_2 = D_4 + D_5 + D_6 + D_7$$

Encoder Example (cont.)

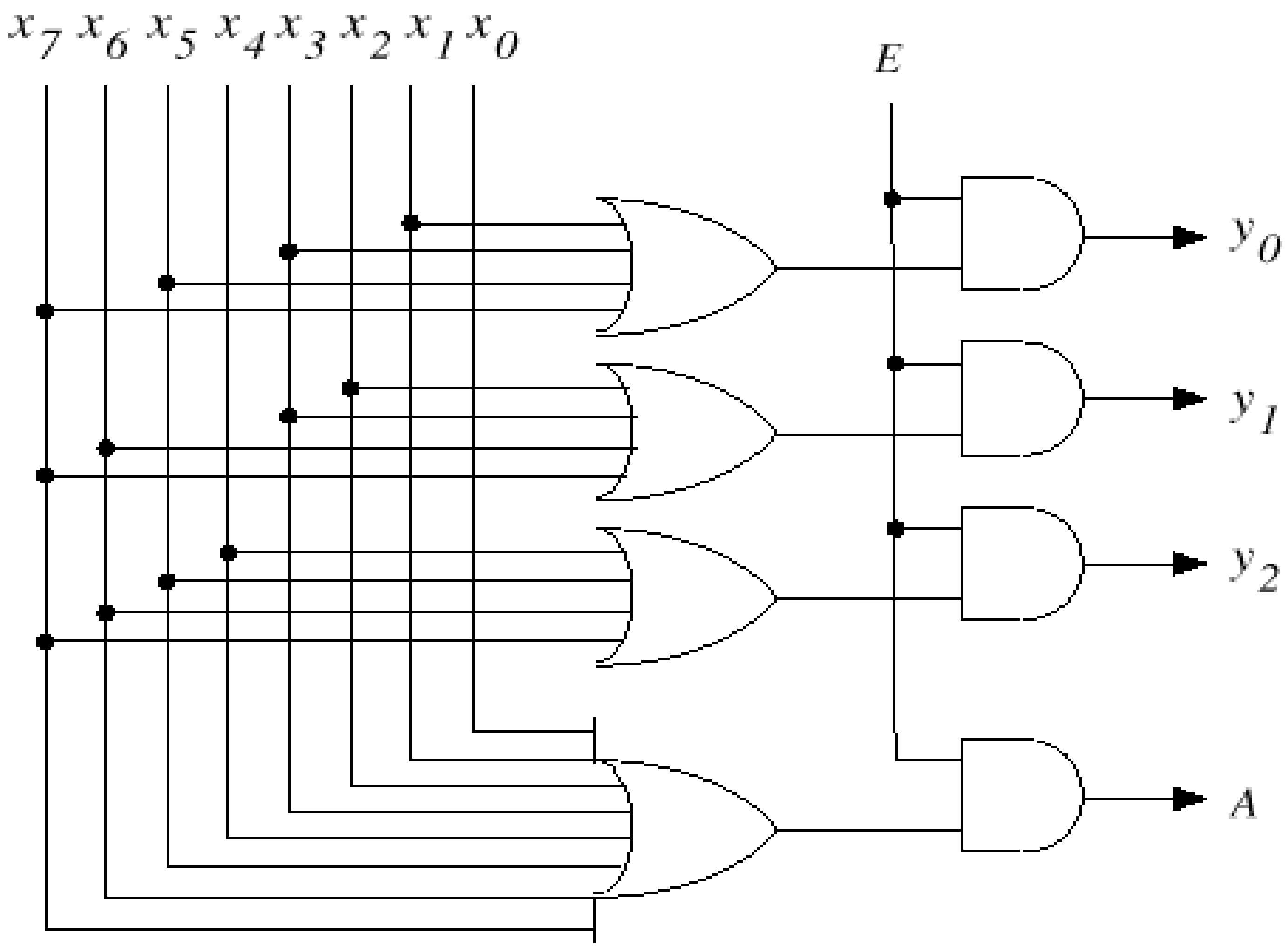


Figure 9.13: mplementation of an 8-input binary encoder.

Encoder Design Issues

- There are two ambiguities associated with the design of a simple encoder:
  1. Only one input can be active at any given time. If two inputs are active simultaneously, the output produces an undefined combination (for example, if D_3 and D_6 are 1 simultaneously, the output of the encoder will be 111.
  2. An output with all 0's can be generated when all the inputs are 0's,or when D_0 is equal to 1.

Priority Encoders

- Solves the ambiguities mentioned above.
- Multiple asserted inputs are allowed; one has priority over all others.
- Separate indication of no asserted inputs.

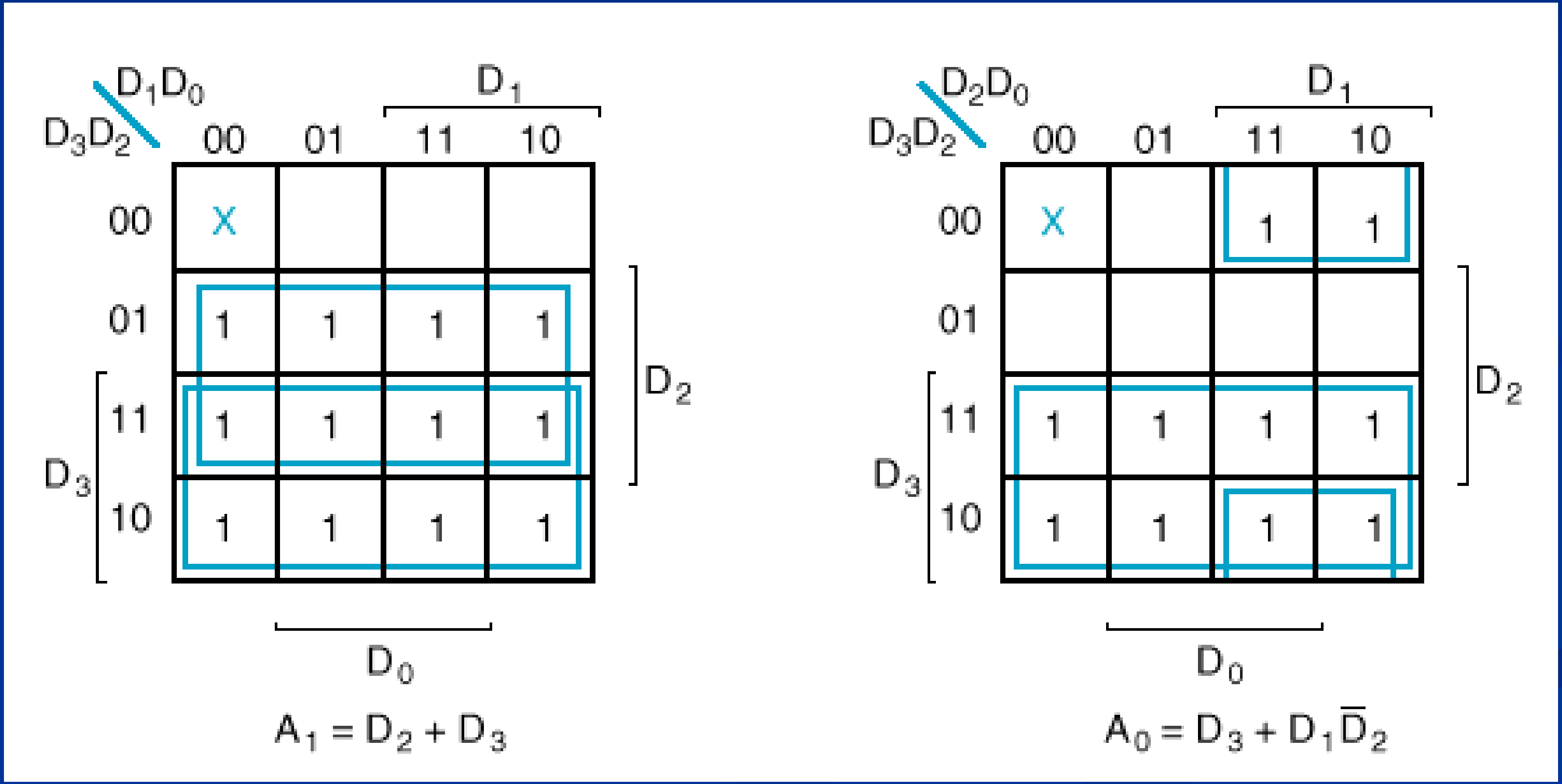
Example: 4-to-2 Priority Encoder Truth Table

Inputs				Outputs		
D_3	D_2	D_1	D_0	A_1	A_0	V
0	0	0	0	X	X	0
0	0	0	1	0	0	1
0	0	1	X	0	1	1
0	1	X	X	1	0	1
1	X	X	X	1	1	1

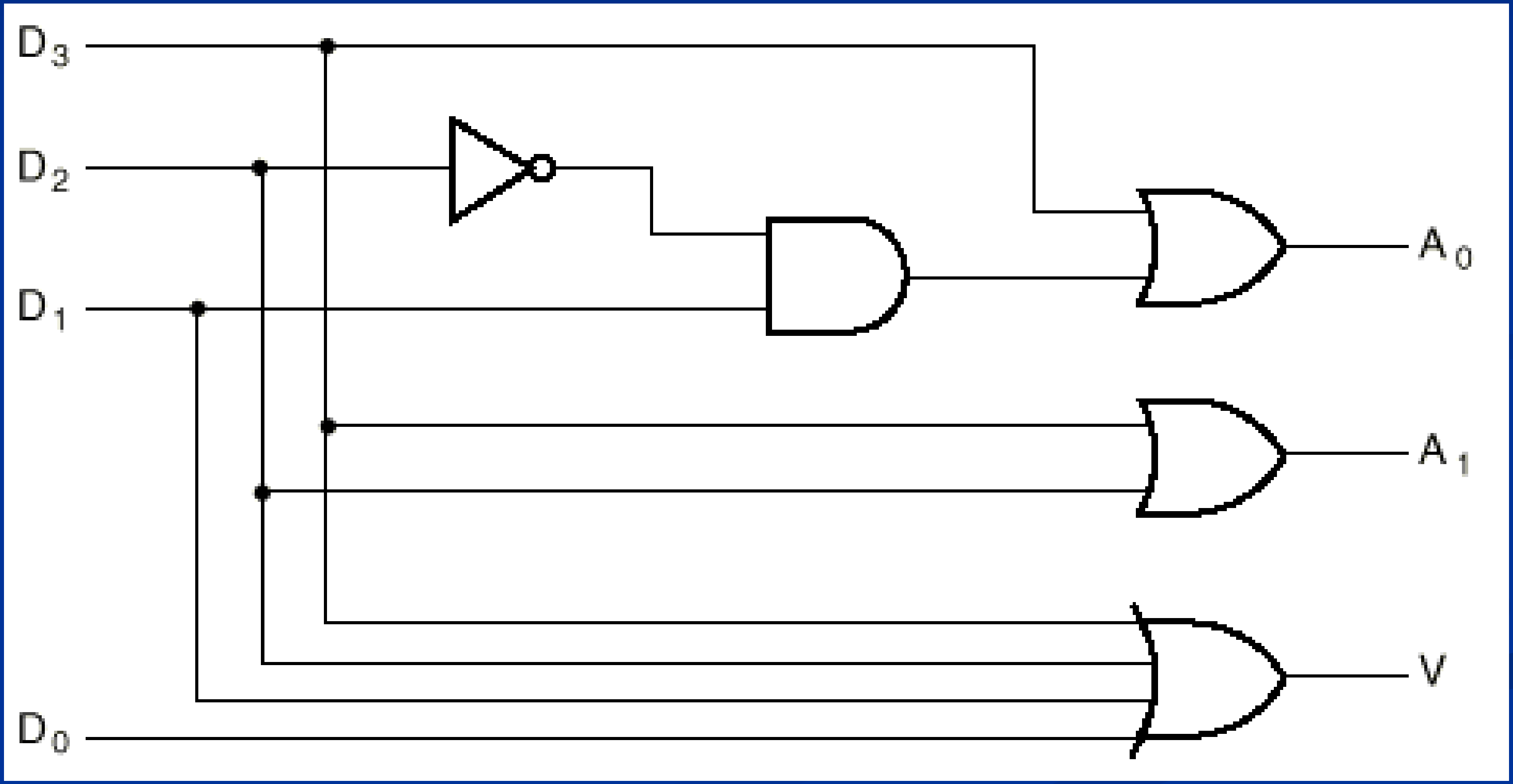
4-to-2 Priority Encoder (cont.)

- The operation of the priority encoder is such that:
- If two or more inputs are equal to 1 at the same time, the input in the highest-numbered position will take precedence.
- *A **valid output indicator***, designated by V, is set to 1 only when one or more inputs are equal to 1. $V = D_3 + D_2 + D_1 + D_0$ by inspection.

Example: 4-to-2 Priority Encoder K-Maps



Example: 4-to-2 Priority Encoder Logic Diagram



8-to-3 Priority Encoder

Table 4.6 A priority encoder.

A_0	A_1	A_2	A_3	A_4	A_5	A_6	A_7	Z_0	Z_1	Z_2	NR
0	0	0	0	0	0	0	0	X	X	X	1
X	X	X	X	X	X	X	1	1	1	1	0
X	X	X	X	X	X	1	0	1	1	0	0
X	X	X	X	X	1	0	0	1	0	1	0
X	X	X	X	1	0	0	0	1	0	0	0
X	X	X	1	0	0	0	0	0	1	1	0
X	X	1	0	0	0	0	0	0	1	0	0
X	1	0	0	0	0	0	0	0	0	1	0
1	0	0	0	0	0	0	0	0	0	0	0

Uses of priority encoders (cont.)

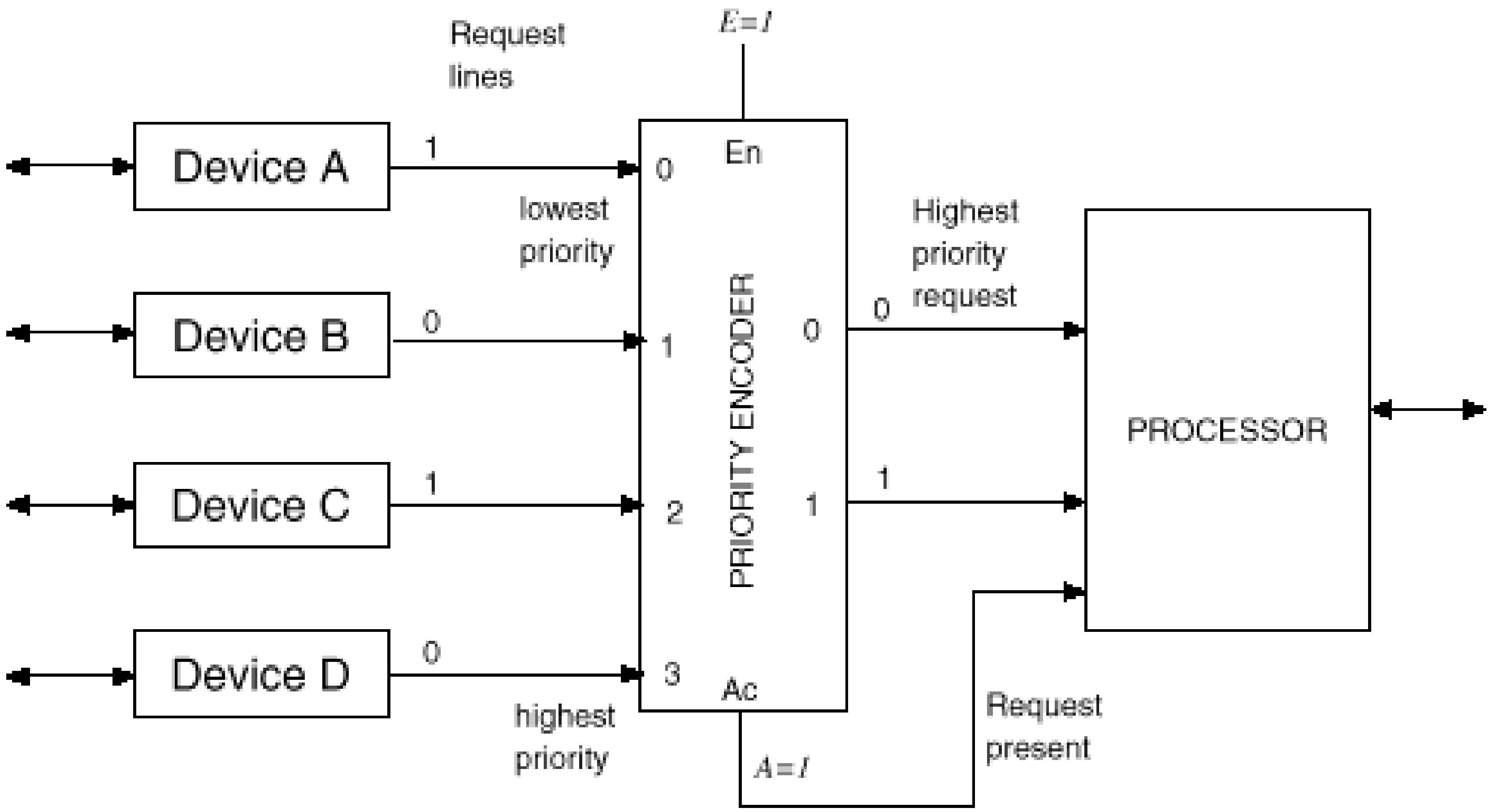
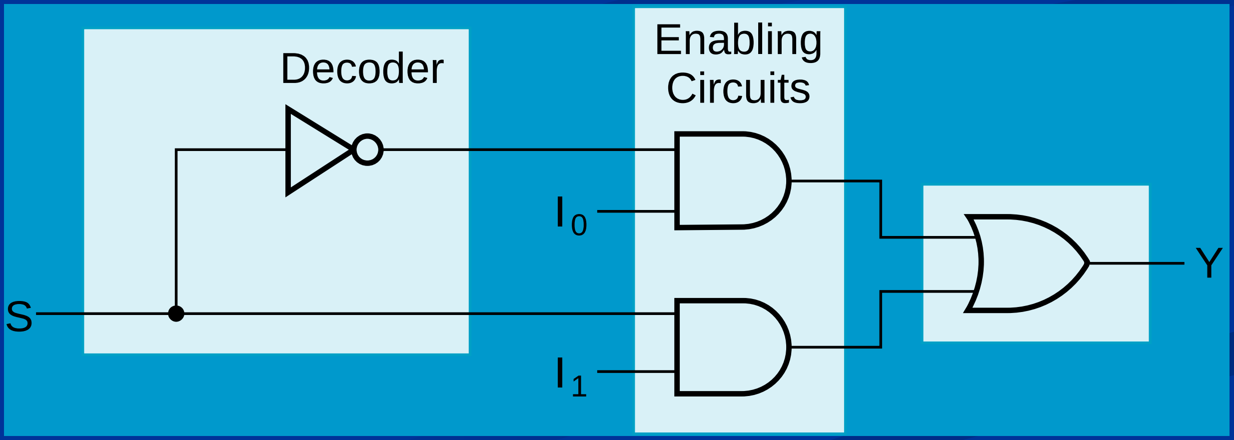


Figure 9.17: Resolving interrupt requests us ng a prio ity encoder.

Link Between Multiplexer and Decoder

- Note the regions of the multiplexer
  - 1-to-2-line Decoder
  - 2 Enabling circuits
  - 2-input OR gate
- In general, for an 2^n-to-1-line multiplexer:
  - n-to-2^n-line decoder
  - 2^n AND gates



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  - Circuit implementation
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